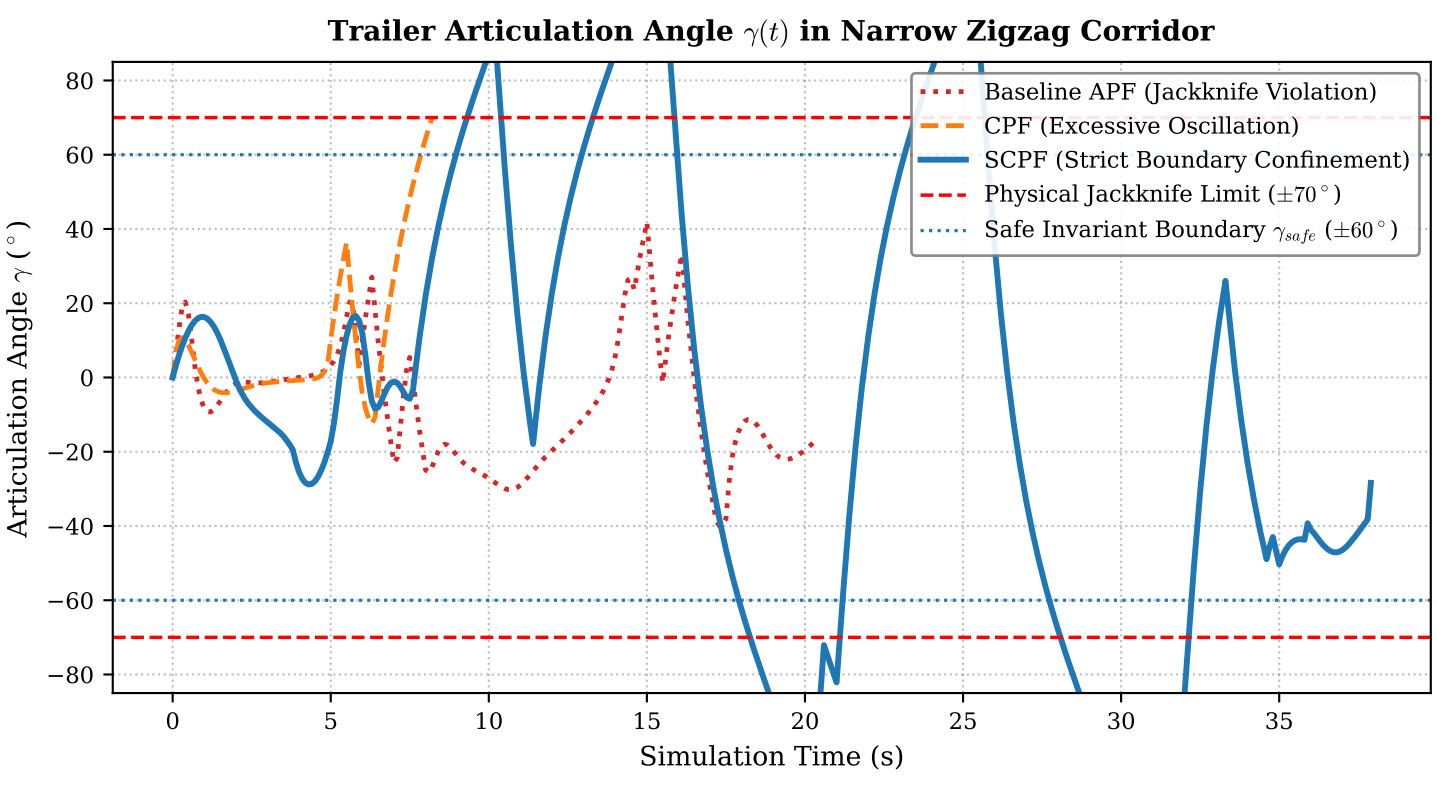
Trailer Articulation Angle $\gamma(t)$ in Narrow Zigzag Corridor
Baseline APF (Jackknife Violation)
CPF (Excessive Oscillation)
SCPF (Strict Boundary Confinement)
Physical Jackknife Limit ($\pm 70^\circ$)
Safe Invariant Boundary γ_{safe} ($\pm 60^\circ$)
Articulation Angle γ ($^\circ$)
Simulation Time (s)
80
60
40
20
0
−20
−40
−60
−80
0
5
10
15
20
25
30
35